

Family 1.0.tf in two pages

the land and its coasts, measured finer than 10km and 5 days

blauewelt/earth — two-page summary of the design note, 20 September 2026. A glossary of every term follows the two pages.

What it is. Family 1.0.tf is a proposed set of datasets (“stores”) that gives the programme’s forecasting model the continents and a 100km coastal band at **10km or finer, every 5 days or finer**: weather stations and balloons, tide gauges, buoys, gliders, flux towers, fire detections, rivers and soil probes; laser measurements of forests from orbit; daily satellite land temperature, snow and surface colour; fluorescence, leaf area and the leaf calendar; three billion species records; a list of every satellite and aerial image of the land since 1982; and the land-change maps the model should learn to predict. **What it is for:** predicting land change — where forest will be cleared, how much carbon a canopy holds and how fast it changes, floods, fire, crops, cities — and giving the four ocean goals (El Niño, the carbon sink, the Atlantic overturning, sea-surface temperature) the land and coast that border the sea. **The land biosphere is the programme’s fifth read-out** beside those four, scored on two held-out measurements: the carbon flux of flux towers the model was not trained on, and canopy height at laser footprints the model was not trained on. **What it is not:** not an archive of pixels — the imagery (petabytes a year) is listed, not copied, until a small learned encoder turns image tiles into tokens; not gap-filled analyses; and not the embedding route, which is family 0.9.tf. Sources whose licences forbid republishing are built on a private track and never redistributed.

store	what it is	producer	space	time	record	stored size	track	phase
<i>point networks: stations, balloons, tide gauges, buoys, gliders, towers, fires, rivers, soil, coastal radar and altimetry</i>								
ghcnd	daily temperature, rain and snow at weather stations	NOAA NCEI (GHCN-Daily)	point; 132,437 stations	daily	1763 →	note $\approx$ 40 GB. Probe (2024-01): 1.02 M station-days, 41 B/row; 12.25 M rows in 2024	public	A
igra	weather-balloon soundings of the air column	NOAA NCEI (IGRA v2.2)	point; 2,932 sites	00 and 12 UTC	1905 →	note 10 GB. Probe (2020-01): 41,921 soundings, 191 B/row $\Rightarrow$ 39–48 M soundings, 7–9 GB	public	A
tide	hourly coastal sea level from tide gauges	GESLA-4 via UHSLC	point; 6,474 records	hourly or finer	1800s → 2026	note 40–50 GB. Probe (2019-03): 2.44 M rows, 33 B/row (80 MB a month)	public	A
tide-private	tide gauges whose terms forbid republishing	GESLA-4 (CMEMS, CV, UZ; SANHO)	point	hourly or finer	as tide	note 10–20 GB [†] . Probe (2019-03): 0.71 M rows, 33 B/row (24 MB a month)	private	A
ndbc	weather and waves at US buoys and coastal stations	NOAA NDBC	point; 1,345 stations	hourly; 10 min recently	1970 → 2025	note 15–25 GB. Probe (2023-06): 3.39 M rows, 47 B/row $\Rightarrow$ $\approx 6.3 \times 10^8$ rows, $\approx$ 30 GB	public	A
gliders	ocean profiles from autonomous underwater gliders	IOOS Glider DAC; OceanGliders	point profile	every $\approx$ 3 h	2003 →	note 0.8 GB. Probe (2023-06): 13,687 profiles, 155 B/row (2.1 MB a month)	public	A
flux	land–air exchange of CO ₂ , water and heat	FLUXNET (AmeriFlux, ICOS, TERN, ChinaFLUX)	point; 700+ towers	half-hourly	1991 →	5 GB	public	A
fire	satellite detections of active fires	NASA FIRMS (MODIS, VIIRS)	1 km / 375 m	per overpass	2000-11 →	20 GB	public	A
river	daily river discharge (US and Caravan)	USGS NWIS; GRDC-Caravan	point; 24,976 + 5,356	daily	1800s →	14 GB	public	B
river_grdc	daily river discharge worldwide	GRDC	point; > 10,000 gauges	daily	1800s →	$\approx$ 5 GB [†]	private	B
ismn	soil moisture and temperature in the ground	ISMN (TU Wien)	point; 3,300 stations	hourly	1952 →	$\approx$ 15 GB [†]	private	B
ghcnh	hourly weather-station reports	NOAA NCEI (GHCNh)	point; 38,870 stations	hourly / synoptic	1700s →	360 GB	public	B
hfr	surface currents from shore-based radar	HFRNet (US); European HFR Node	6 / 2 / 1 km cells	hourly	US 2012 →; EU $\approx$ 2009 →	110–155 GB	public	B
alticap	satellite sea level close to the coast	ALTICAP (AVISO/ODATIS), Jason-3	$\approx$ 350 m along track	$\approx$ 10-day repeat	2016-02 → 2021-07	3 GB	public	B
ecad	European station weather missing from GHCN-Daily	ECA&D (KNMI)	point	daily	1800s →	$\leq$ 20 GB [†]	private	C
<i>LiDAR: laser heights of forests, ground and sea ice from orbit</i>								
gedi	laser canopy height, cover and biomass	NASA GEDI (LP DAAC, ORNL)	25 m footprints	per orbit pass	2019-04 → 2023-03; 2024-04 → 2025-07	0.2–0.5 TB for the record	public	A (one year, 2022) / B

store	what it is	producer	space	time	record	stored size	track	phase
icesat2	laser terrain, canopy and sea-ice height	NASA ICESat-2 (NSIDC)	100 m segments (≈ 11 m spots)	per orbit pass	2018-10 $\rightarrow$	≈ 1.1 TB for the record	public	A (one year, 2022) / B
<i>the biosphere: photosynthesis, leaf area and its calendar, canopy and biomass maps, species records</i> canopy30	how tall the canopy is, one map at 30 m	UMD GLAD (forest height 2020)	30 m	one year	2020	probe: ≈ 33 GB (tile 50N_000E: 281 MB stored, dense); 261 tiles, 35.95 GB native, measured	public	A (a box: one bin, 11–21 h)
canopy_ref	the finer canopy maps, listed not copied	ETH Zurich (10 m); Meta/WRI (1 m, on AWS)	10 m and 1 m	annual	2020 (ETH); 2018–2020 (Meta)	catalogue < 1 GB	public	A
sif	fluorescence: photosynthesis seen from orbit	TROPOMI (Caltech; ESA S5P-PAL); NASA OCO-2, OCO-3	7×3.5 km; OCO 1.3×2.25 km	per sounding	2018-05 $\rightarrow$; OCO-2 2014-09 $\rightarrow$	probe (2023-07): 46.8 GB a year uncut, 101.7 M rows a month	public	A
lai500 (E7)	leaf area, and the light the canopy absorbs	NASA MODIS Terra and Aqua, VIIRS	500 m	8-day composite (E7)	Terra 2000-02 $\rightarrow$; VIIRS 2012 $\rightarrow$	≈ 60 GB [†] per sensor-year	public	A (Terra 2020–2024) / B
pheno500	the leaf calendar: green-up, peak, dormancy	NASA MODIS (MCD12Q2 v061)	500 m	annual	2001 $\rightarrow$ 2025	≈ 10 GB [†] a year	public	A
gbif	dated, located records of species	GBIF snapshot (AWS Open Data)	point; each row's own uncertainty	per record, day or month	1600s $\rightarrow$; 3.67 B records measured	≈ 90 GB [†] public; the 2026-09 snapshot is 285.3 GB	public (83.2 %); private (CC BY-NC, 16.8 %)	A if public, else C
cat.biomass_esa	list of ESA's P-band radar products	ESA BIOMASS (launched 2025-04-29)	—	per product	2025 $\rightarrow$	catalogue	terms unverified	A
<i>daily fields on a 0.05° (5.6 km) grid</i> lst05	land-surface temperature, day and night (Terra in phase A)	NASA MODIS Terra (+ Aqua, VIIRS)	5.6 km (0.05°)	daily	2000-02 $\rightarrow$	≈ 350 GB per satellite	public	A / B
snow05	snow-cover fraction	NASA MODIS (NSIDC)	0.05°	daily	2000-02 $\rightarrow$	≈ 60 GB	public	A
refl05	surface reflectance: the colour of the ground	NASA MODIS (+ VIIRS)	0.05°	daily	2000-02 $\rightarrow$	≈ 700 GB (+330 GB VIIRS)	public	B
ims1k	analyst-drawn snow and ice chart	NOAA IMS (NSIDC)	1 km (4 km and 24 km earlier)	daily	1 km 2014-12 $\rightarrow$; 4 km 2004, 24 km 1997	≈ 50 GB	public	B
chirps05 (E4)	rainfall from satellite infrared and gauges	UCSB CHC (CHIRPS v3.0)	0.05°	pentad (6 a month)	1981 $\rightarrow$	≈ 40 GB	public	B
<i>image catalogue (tier T): every scene listed, pixels left at the source; the ten catalogues together $\approx 1.6 \times 10^8$ rows, ≈ 7 GB</i> cat.landsat	list of every Landsat scene, all sensors	USGS	30 m (thermal 60–120 m)	$\approx 1,400$ – $1,500$ scenes a day [†]	1982 $\rightarrow$	catalogue	public	A
cat.hls	list of harmonised Landsat and Sentinel-2 scenes	NASA LP DAAC (HLS)	30 m	2.71 M + 1.75 M granules in 2025	2013-04 / 2015-11 $\rightarrow$	catalogue	public	A
cat.s2	list of Sentinel-2 optical scenes	ESA / CDSE	10 / 20 / 60 m	12.7–15.5 k scenes a day	2015-07 $\rightarrow$	catalogue	public	A
cat.s1	list of Sentinel-1 C-band radar scenes	ESA / CDSE	10 m	982 scenes a day	2014-10 $\rightarrow$	catalogue	public	A
cat.nisar	list of NISAR L-band radar scenes	NASA-ISRO (ASF)	10 / 20 / 80 m	28,456 granules in Aug 2026	2026-06 $\rightarrow$	catalogue	public	A
cat.palsar (E5)	list of Japanese L-band radar mosaics and scenes	JAXA	25 m	yearly mosaics and scenes	2007–2010, 2015–2025; 2014-08 $\rightarrow$	catalogue	public	A, once a JAXA account exists
cat.olci	list of 300 m coastal ocean-colour scenes	Copernicus Sentinel-3 OLCI	300 m	457 scenes a day	2016-04 $\rightarrow$	catalogue	public	A
cat.viirs	list of VIIRS imager scenes	NASA LAADS	375 m	225–242 granules a day	2012 / 2018 $\rightarrow$	catalogue	public	A
cat.ecostress	list of thermal images from the Space Station	NASA ECOSTRESS	70 m	irregular	2018-07 $\rightarrow$	catalogue	public	A
cat.submetre (E6)	list of sub-metre aerial photos where published	USDA NAIP; swisstopo; Planet-NICFI	0.1–4.77 m	2–3-yearly; NICFI monthly	2009 / 2017 / 2015 $\rightarrow$	catalogue	public; NICFI private	A
<i>read-out targets: what the model is asked to predict</i> alerts	dated forest-clearing alerts	UMD GLAD, WUR RADD, NASA OPERA	10–30 m	per detection	2015 $\rightarrow$	≈ 30 GB	public	A
lossyear	year each 30 m pixel lost its forest	UMD Hansen GFC; EU JRC TMF	30 m	annual	1990 / 2001 $\rightarrow$ 2025	≈ 30 GB	public	A
burned500	date each pixel burned	NASA MODIS MCD64A1	500 m	monthly	2000-11 $\rightarrow$	≈ 10 GB	public	A

store	what it is	producer	space	time	record	stored size	track	phase
cat_gfm	list of flood and surface-water maps	Copernicus GFM; NASA OPERA DSWx	20–30 m	per pass	2015 / 2016 →	< 1 GB	public	A
biomass100	yearly above-ground biomass	ESA CCI Biomass; GEDI L4B	100 m (GEDI 1 km)	annual	2005–2012, 2015–2024; L4B 2019–2023	≈ 0.4 TB	public (terms read 2026-09-20)	A (2020 and all of L4B) / B
landcover10	yearly land-cover and crop-type maps	Esri/IO, ESA, MapBiomass, USDA, JRC	10–30 m	annual	1985 / 2017 →	≈ 0.2 TB [†]	public	C
statics								
static_fine	fixed maps: elevation and depth, land cover, distance to coast	GEBCO, ESA WorldCover, GSHHG	15'' (≈ 460 m); 0.05°	fixed	—	≈ 8 GB	public	A
<i>phase-C additions made from the imagery</i>								
oc300_coast	300 m coastal ocean colour as a grid	Copernicus GlobColour (OLCI)	300 m	gridded level-3 frames	2016-04 →	≈ 1.6 TB	public	C
tile tokens	codec vectors for each image tile	this project (E-078c)	10–30 m tiles	per scene	—	measured by E-078c	public	C

Sizes are the note's estimates unless marked **Probe**: a one-month test fetch through the real code (E-082, 17 Sep 2026), giving rows and stored bytes per row and, where stated, the whole store scaled from it. [†] estimate. B = bytes. Already held and not repeated: family 7.2's 1° land channels, family 10.1's moorings and along-track sea level, family 10.2's fishing effort.

How it is stored. The same design as family 1.gf: a registry listing every store; *point stores* as column arrays sorted by five-day bin (64-bit time for records before 1914), with a `platforms.json` that keeps each station's elevation; *daily fields* as compressed 256×256 tiles per five-day bin, read a few tiles at a time; *catalogues* as one row per scene (time, centre, cloud and valid fraction, sun or radar angle, area, a sensor code) with a side table of download links. A LiDAR profile's heights and a sounding's levels are channels. A yearly target is filed at the end of its year, so it is never an input inside that year. The coastal band (± 100 km) is a recorded build setting. Private stores (`tide_private`, `river_grdc`, `ismn`, `ecad`; NICFI) sit in a private repository and are listed publicly.

The ladder

- A** ≈ 0.7 TB, under \$10: the point stores `ghcnd`, `igra`, `tide` + `tide_private`, `ndbc`, `gliders`, `flux`, `fire`; Terra `1st05` and `snow05`; all ten catalogues (`cat_palsar` once a JAXA account exists); targets `alerts`, `lossyear`, `burned500`, `cat_gfm`; `static_fine`. Six of the point stores have run their one-month probes. *Plus the biosphere wave*, ≈ 1 TB[†]: `canopy30`, `canopy_ref`, `sif` (TROPOMI), `pheno500`, `gbif`, `cat_biomass_esa`, `biomass100` for 2020 and all of GEDI L4B, and one year each of `gedi`, `icesat2` and `lai500` (Terra 2020–2024) — every new store probed before it is built.
- B** 2.3–2.7 TB, \$20–30: `ghcnh`, `hfr`, `river`, `river_grdc`, `ismn`, `alticap`, `ref105`, `ims1k`, `chirps05`, the Aqua and VIIRS continuations; and, on the probes' measured numbers, the rest of `gedi`, `icesat2`, `lai500` and `biomass100`.
- C** ≈ 3 TB, after the image-codec experiment E-078c: `landcover10`, `ecad`, the tile tokens, `oc300_coast`.

Glossary

Every term, acronym, producer, instrument and code name used in this summary and in the full design note, in alphabetical order.

alerts — Family 1.0.tf target store of dated forest-disturbance alerts (GLAD, RADD, OPERA) as point rows.

ALOS, PALSAR, PALSAR-2 — Japan’s Advanced Land Observing Satellites and their L-band radars.

AlphaEarth Foundations (AEF) — Google DeepMind’s model that turns sequences of satellite images into one 64-number description of each 10 m of land for each year; its published output is the *Satellite Embedding*.

ALTICAP — A coastal altimetry product that reprocesses Jason-3 to within a few hundred metres of the shore.

alticap — Family 1.0.tf store of ALTICAP coastal sea level along Jason-3 tracks, 2016–2021.

altimeter, along-track — A radar satellite that measures sea-surface height straight below it; along-track data are its measurements along that ground line.

AmeriFlux, ICOS, TERN, ChinaFLUX, FLUXNET — The flux-tower networks of the Americas, Europe, Australia and China, and FLUXNET, their global collection (the FLUXNET Shuttle release).

AMOC — The Atlantic Meridional Overturning Circulation: the system of Atlantic currents that carries warm water north near the surface and cold water south at depth, one of the programme’s four prediction goals.

anchor — The place and time the model is asked to predict; everything it reads is placed relative to it.

annual target — A yearly map the model is asked to predict (a loss year, a biomass map); it is filed at the end of its year so it is never read as an input from inside that year.

Argo, BGC-Argo, Sprof — The global array of about 4,000 drifting floats that profile temperature and salinity to 2,000 m; BGC-Argo floats add chemistry sensors; Sprof files merge each float’s variables.

ASF — The Alaska Satellite Facility, NASA’s archive for radar data including NISAR.

assets.parquet — A table beside each image catalogue mapping every scene to its catalogue name and download links.

ATLAS; ATL07, ATL08, ATL10 — ICESat-2’s photon-counting laser, and its products for sea-ice height, land-and-vegetation height, and sea-ice freeboard.

attribution, yes, no (licence.redistribution) — Registry values saying whether a licence lets the data be republished freely, with credit, or not at all.

AVISO, ODATIS — French services that distribute satellite altimetry and ocean data.

AWS, AWS Open Data — Amazon Web Services, and its programme that hosts public datasets at no cost to readers.

bathymetry — The depth of the sea floor.

bin (pentad) — The five-day time step every family uses, counted from 1 January 1982; a bin before 1982 has a negative number.

BIOMASS (the satellite), P band — ESA’s radar satellite, launched 29 April 2025; its 70 cm (P band) wavelength reaches through a canopy to the trunks, which is why it measures woody biomass directly. Its products are listed in `cat_biomass_esa`.

biomass100 — Family 1.0.tf target store of yearly above-ground biomass maps at 100 m.

box (rented, verified) — A machine hired by the hour from a marketplace; it holds no credentials, and a verified one runs on a host the marketplace has vetted.

B/row (bytes per row) — Stored size of one row of a point store; a row costs 27 + 2 bytes per channel (31 + 2 per channel with the 64-bit time).

burned500 — Family 1.0.tf target store of the date each 500 m pixel burned.

canopy30 — Family 1.0.tf target store: GLAD’s 2020 forest-height map at 30 m, 261 tiles, the canopy read-out’s annual target.

canopy_ref — Family 1.0.tf catalogue of the finer canopy-height maps — ETH Zurich’s 10 m and Meta/WRI’s 1 m — listed with their tile footprints and download links, pixels left at the source.

Caravan — An open (CC BY 4.0) collection of catchment discharge and weather data, including a GRDC subset.

cat_* — The family 1.0.tf image-catalogue stores: one row per satellite or aerial scene (time, place, cloud and valid fraction, sun or radar angle, area, sensor code), without the pixels.

cat_biomass_esa — Family 1.0.tf catalogue of the released products of ESA’s BIOMASS radar satellite.

cat_ecostress — Catalogue of ECOSTRESS 70 m thermal scenes.

cat_gfm — Target catalogue of flood and surface-water maps (Copernicus GFM, OPERA DSWx); a read-out, not imagery.

cat_hls — Catalogue of NASA’s harmonised Landsat and Sentinel-2 scenes.

cat_landsat — Catalogue of every Landsat scene since 1982, all sensors.

cat_nisar — Catalogue of NISAR L-band radar scenes.

cat_olci — Catalogue of Sentinel-3 OLCI 300 m ocean-colour scenes over the coastal band.

cat_palsar — Catalogue of JAXA’s PALSAR yearly radar mosaics and wide-swath scenes (exception E5).

cat_s1 — Catalogue of Sentinel-1 radar scenes.

cat_s2 — Catalogue of Sentinel-2 optical scenes.

cat_submetre — Catalogue of sub-metre and metre-scale aerial and basemap imagery where it is openly published (exception E6).

cat.viirs — Catalogue of VIIRS 375 m imager scenes.

C band, L band — Radar wavelengths of about 5.5 cm and 24 cm; L band sees deeper into vegetation.

CC0, CC BY, CC BY-NC — Creative Commons licences: public domain; free with attribution; free with attribution for non-commercial use only.

C (channels) — The number of values stored per row or per pixel.

CDL — The USDA Cropland Data Layer, a yearly 30 m US crop-type map.

CDSE — The Copernicus Data Space Ecosystem, the archive and catalogue of the Sentinel satellites (free account required).

CEDA — The UK Centre for Environmental Data Analysis, an open archive.

checksum (sha256) — A 64-character fingerprint of a file; if one byte changes, the fingerprint changes.

CHIRPS — The Climate Hazards Center InfraRed Precipitation with Station data rainfall record (from 1981).

chirps05 — Family 1.0.tf store of CHIRPS five-day rainfall at 0.05° (exception E4), with its **frame.table** of frame dates.

climatology — The long-term average for a place and season — the simplest forecast a model has to beat.

C-MAN — NOAA’s Coastal-Marine Automated Network of shore weather stations.

CMEMS, CV, UZ — Three GESLA contributor codes whose tide-gauge data are for research use only.

codec (local codec) — A small learned encoder that turns an image tile into a few compact vectors (tokens) the model can read.

cone — The set of past measurements the model reads around the anchor: wider for older times, like a cone in space and time.

CO-OPS — NOAA’s Center for Operational Oceanographic Products and Services, which runs the US tide gauges.

Copernicus, Copernicus Marine — The EU’s Earth-observation programme and its marine service (free account required).

Copernicus DEM, GLO-30 — The EU’s global 30 m elevation model.

DAAC (LP, ORNL, PO, OB) — NASA’s discipline data centres: LP DAAC for land, ORNL DAAC for ecology, PO.DAAC for oceans, OB.DAAC for ocean biology.

DAC, GDAC — Data assembly centre, and the global one that gathers a network’s data (Argo, gliders, OceanSITES).

degraded — Registry list of every check a build had to relax, so a relaxation can never be silent.

DIST-ALERT — OPERA’s 30 m alerts of vegetation disturbance from HLS.

dist_coast — The 0.05° static field of distance to the nearest coast, from GSHHG.

distribution — Registry field naming a store’s track, **public** or **private**.

DSWx-HLS, DSWx-S1 — OPERA’s 30 m surface-water extent maps from HLS optical and from Sentinel-1 radar.

Dynamic World — Google’s 10 m land-cover classification of each Sentinel-2 scene, available only inside Earth Engine.

E-078, E-078c — The design study that gave family 10 one schema for every granularity, and its first experiment training a local image codec on about 300 coastal cells.

E-081 — The experiment that built family 10.2’s fishing-effort store and its globe layer.

E-082 — The build plan for families 1.gf, 1.0.tf and 0.9.tf; its first wave wrote and probed the phase-A point-store adapters, and its sixth adds the biosphere stores.

E1–E7 (exceptions) — Sources admitted although they break the 10 km / 5-day rule: E1 scatterometer wind at 12.5 km, E2 IMERG rain at 11 km, E3 GOSAT at 10.5 km, E4 CHIRPS rain, E5 PALSAR yearly mosaics, E6 sub-metre photos, E7 the eight-day leaf-area composite **lai500**.

Earthdata login — NASA’s free user account, needed to download most NASA data.

Earth Engine — Google’s hosted geospatial analysis platform (account required).

ecad — Private family 1.0.tf store of ECA&D European station series missing from GHCN-Daily.

ECA&D; TX, TN, TG, RR, SS, SD, HU, PP — The European Climate Assessment & Dataset of station series, and its element codes: maximum, minimum and mean temperature, rain, sunshine, snow depth, humidity and pressure.

ECOSTRESS; LSTE, L2T — A thermal radiometer on the International Space Station; LSTE is its land-surface temperature and emissivity product, L2T the tiled form.

eddy covariance — Measuring the exchange of gases and heat between land and air from fast wind and concentration readings on a tower.

embedding — A learned list of numbers summarising what was observed at a place, arranged so that similar places get similar numbers.

EOSDIS — NASA’s Earth Observing System Data and Information System, the umbrella of its archives.

ERDDAP — A data server that answers subset queries over HTTP.

ESA — The European Space Agency.

ESA CCI — ESA’s Climate Change Initiative, which builds long consistent climate records from satellites.

Esri, Impact Observatory — The companies that produce a yearly 10 m global land-cover map.

EUCROPMAP — The JRC’s crop-type map of the European Union.

EVI — The enhanced vegetation index, a greenness measure computed from red, near-infrared and blue reflectance; the phenology dates of **pheno500** are read off its yearly curve.

falsifier — The test, fixed before building, that can show a store is not worth keeping.

family — A numbered set of datasets built to one storage design and read by the programme’s forecasting model.

family 0.9.tf — Family 1.0.tf with its raw-imagery catalogue replaced by Google’s AlphaEarth embedding, everything else inherited.

family 10, 10.1, 10.2 — The granularity-aware dataset set in which every measurement keeps its own place, time and footprint; 10.1 built four point stores (drifters, tropical moorings, ship CO₂, satellite sea level) and 10.2 inherited them and added fishing effort.

family 1.0.tf — The proposed land-and-coast family at the same scale, built from raw observations and a catalogue of raw imagery (“tf”: terrestrial, fine).

family 1.gf — The proposed global family of ocean and atmosphere measurements finer than 10 km and 5 days (“gf”: global, fine).

family 7, 7.1, 7.2 — The published global input tensor: every 0.25° cell from pole to pole in five-day steps, 1982–2024; 7.1 added ocean colour and 7.2 is the current version.

fire — Family 1.0.tf store of FIRMS satellite fire detections.

FIRMS; FRP — NASA’s Fire Information for Resource Management System, and fire radiative power, the energy a fire gives off.

flux — Family 1.0.tf store of half-hourly flux-tower measurements.

footprint; $\log_2 f_p$, $\log_2 f_t$ (log2_fp, log2_dt, fp) — The area and time span a value averages over, stored as base-2 logarithms relative to 27.83 km and 5 days; -4 is the label for a point measurement with no area.

forcing — An input that drives the system being predicted, such as wind over the ocean or rain over land.

FPAR — The fraction of photosynthetically active radiation a canopy absorbs — the share of usable sunlight the leaves catch, a channel of lai500.

frame; frames_per_bin, frame.seconds, F — One time slice of a gridded field; a five-day bin holds F of them (5 for daily, 40 for three-hourly), each lasting frame.seconds.

freeboard — The height of sea ice above the waterline, from which its thickness is estimated.

gbif — Family 1.0.tf store of dated, located species-occurrence records, built from GBIF’s monthly snapshot; the records whose licence forbids commercial use go to the private track.

GBIF — The Global Biodiversity Information Facility, the pooled open archive of species-occurrence records (about four billion); *taxon key* is its identifier for a species or higher group and *basis of record* says how an occurrence was observed — a specimen, a person, or a machine.

GEBCO — The General Bathymetric Chart of the Oceans, a global elevation and sea-floor-depth grid.

gedi — Family 1.0.tf store of GEDI laser footprints: canopy height profile, cover and biomass.

GEDI; L2A, L2B, L4A, L4B — The Global Ecosystem Dynamics Investigation, a laser on the Space Station measuring canopy height; its products for height profiles, cover, footprint biomass and gridded biomass.

GES DISC — NASA’s Earth-science data centre for the atmosphere; it serves the OCO fluorescence and column-CO₂ products and needs its own approval on an Earthdata account.

GESLA, GESLA-4 — The Global Extreme Sea Level Analysis compilation of hourly tide-gauge records, version 4.

GFM — Copernicus’ Global Flood Monitoring service, flood extent from every Sentinel-1 pass.

ghcnd — Family 1.0.tf store of GHCN-Daily station records.

GHCN-Daily, GHCNh, ISD — NOAA’s Global Historical Climatology Network of daily and hourly station records; GHCNh replaces the frozen Integrated Surface Database.

ghcnh — Family 1.0.tf store of GHCN-hourly station reports.

GHSL — The EU’s Global Human Settlement Layer of built-up areas.

GitHub, hosted runner — The code-hosting service the project uses; a hosted runner is one of GitHub’s own build machines, the only place credentials are used.

GLAD, GLAD-L, GLAD-S2 — The University of Maryland’s Global Land Analysis and Discovery laboratory, and its forest-loss alerts from Landsat and from Sentinel-2; it also publishes the Hansen loss maps and the 2020 forest-height map behind canopy30.

gliders — Family 1.0.tf store of glider profiles at 16 depths.

GlobColour — Copernicus’ merged ocean-colour product.

GPP, NEE, Reco, LE, H, Rn, Ta, VPD, SWC, P — Flux-tower variables: photosynthesis, net ecosystem exchange of CO₂, ecosystem respiration, latent and sensible heat, net radiation, air temperature, vapour-pressure deficit, soil water content, precipitation.

GRDC — The Global Runoff Data Centre’s archive of river discharge from more than 10,000 gauges.

GSHHG — The Global Self-consistent Hierarchical High-resolution Geography shoreline set (public domain).

gtmba — Family 10.1’s store of the tropical moored buoy arrays, including TAO.

HadISD — The Met Office’s quality-controlled hourly station set, a derivative of ISD.

Hansen GFC — The University of Maryland’s Global Forest Change map of tree cover in 2000 and the year of loss since.

hfr — Family 1.0.tf store of hourly surface currents from shore-based radar.

HFR, HFRNet — High-frequency radar that measures surface currents from the shore, and the US network of such radars

(with a European node).

HLS; S30, L30 — NASA’s Harmonized Landsat Sentinel-2 30 m product, from Sentinel-2 (S30) and Landsat (L30).

HTTP, HTTPS, HTTP range — The web’s transfer protocol (HTTPS is the encrypted form); a range request asks for one byte span of a file.

Hub (Hugging Face) — Hugging Face’s dataset hosting, where the project publishes its stores; it limits files per repository and per folder.

ICESat-2 — NASA’s polar laser-altimetry satellite (2018 on).

icesat2 — Family 1.0.tf store of ICESat-2 terrain, canopy and sea-ice heights per 100 m segment.

IGRA — The Integrated Global Radiosonde Archive of weather-balloon soundings.

igra — Family 1.0.tf store of IGRA radiosonde soundings at 16 standard levels.

IMS — NOAA’s Interactive Multisensor Snow and Ice Mapping System, a daily analyst-drawn chart.

ims1k — Family 1.0.tf store of the daily 1 km IMS snow and ice chart.

IOOS — The US Integrated Ocean Observing System, which runs the national glider data centre.

ISMN — The International Soil Moisture Network of in-ground sensors.

ismn — Private family 1.0.tf store of ISMN soil moisture and soil temperature.

ISRO — The Indian Space Research Organisation, NASA’s partner on NISAR.

ISS — The International Space Station, platform of GEDI and ECOSTRESS.

Jason-3 — A radar altimeter satellite (2016 on) measuring sea level.

JAXA — Japan’s space agency.

JRC — The European Commission’s Joint Research Centre.

JRC TMF — The JRC’s Tropical Moist Forest map of yearly forest change since 1990.

JSON — A plain-text format for structured data, used for every registry.

KB, MB, GB, TB, PB — Kilo-, mega-, giga-, tera- and petabytes: each a thousand times the last.

Kelvin wave — An equatorial ocean wave that carries a bump of sea level eastward across the Pacific in about two months, a precursor of El Niño.

KNMI — The Royal Netherlands Meteorological Institute, which runs ECA&D.

LAADS — NASA’s Level-1 and Atmosphere Archive and Distribution System.

lai500 — Family 1.0.tf store of eight-day leaf area and FPAR at 500 m from MODIS and VIIRS (exception E7).

landcover10 — Family 1.0.tf target store of yearly land-cover and crop-type maps.

Landsat 4, 5, 7, 8, 9; TM, ETM+, OLI, TIRS — The US optical Earth-observation satellites since 1982, and their cameras: Thematic Mapper, Enhanced Thematic Mapper Plus, Operational Land Imager and Thermal Infrared Sensor; Collection 2 is the current reprocessing.

lat, lon, values, qc, platform, time_s — Arrays of a point store: position, the measured values, a quality code, a hashed instrument identity, and the time in seconds since 1982.

leaf-area index (LAI) — The area of leaf above a square metre of ground; two and a half means two and a half square metres of leaf over each square metre of land.

level 1B, 2, 3, 3C, 3S, 4 (L1B, L2, L3, L3C, L3S, L4) — Processing stages of satellite data: calibrated signals; measurements on the instrument’s own swath; measurements binned onto a grid; binned per sensor; merged across sensors; and gap-filled analyses, which are derived.

LiDAR — Laser ranging: a pulse’s return time gives the height of the ground and of each layer of a canopy.

lossyear — Family 1.0.tf target store of the year each 30 m pixel lost its forest.

1st05 — Family 1.0.tf store of daily 0.05° land-surface temperature, day and night.

MapBiomass — Brazil’s annual 30 m land-cover mapping project.

masked-token loss — The model’s error when asked to predict tokens hidden from it; lower means the inputs helped.

MOD09CMG, MOD10C1, MOD11C1, MYD11C1, MCD64A1 — MODIS product names: daily 0.05° reflectance, snow cover and land-surface temperature (MOD for Terra, MYD for Aqua), and monthly burned area (MCD, both).

MOD15A2H, MYD15A2H, VNP15A2H; MCD12Q2 — More MODIS and VIIRS product names: the eight-day 500 m leaf-area and FPAR products from Terra, Aqua and VIIRS, and the yearly 500 m land-surface phenology product behind pheno500.

MODIS; Terra, Aqua — The Moderate Resolution Imaging Spectroradiometer on NASA’s Terra (1999) and Aqua (2002) satellites, both near the end of their lives.

MSWEP — A merged global precipitation product (non-commercial licence).

NAIP — The USDA’s National Agriculture Imagery Program of 0.6–1 m aerial photographs.

NaN — “Not a number”, the stored value for “not measured”.

NASA — The US space agency.

NCEI, OCADS — NOAA’s National Centers for Environmental Information, the US national environmental archive, and its ocean-carbon archive.

NDBC — NOAA’s National Data Buoy Center.

ndbc — Family 1.0.tf store of NDBC buoy and coastal-station weather and waves.

NICFI, Planet-NICFI — Norway’s International Climate and Forests Initiative, which paid for Planet’s free monthly tropical basemaps.

Niño-3.4, Niño indices — The average SST of fixed boxes in the equatorial Pacific; Niño-3.4 is the standard El Niño index.

NISAR; GCOV — The NASA-ISRO L-band radar satellite (launched 2025), and its geocoded covariance product.

NOAA, NOAA STAR — The US National Oceanic and Atmospheric Administration, and its satellite research centre.

n_R — The number of readings a token carries; $n_R = 0$ says “nothing was measured here”.

NSIDC — The US National Snow and Ice Data Center.

NWIS — The USGS National Water Information System, the US river-gauge archive.

O₂, NO₃, PO₄, SiO₄, pH — Dissolved oxygen, nitrate, phosphate and silicate, and the acidity of seawater.

oc300_coast — A possible phase-C gridded store of 300 m OLCI ocean colour over the coastal band.

OceanGliders — The international glider programme and its data centre.

OCO-2, OCO-3; SIF Lite — NASA’s two Orbiting Carbon Observatories and their per-sounding fluorescence files, the second and third platforms of **sif**.

OGD — Open government data licence.

OLCI; WFR — Sentinel-3’s Ocean and Land Colour Instrument, and its full-resolution water product.

OPERA — NASA’s Observational Products for End-Users from Remote Sensing Analysis: disturbance and surface-water maps from HLS and Sentinel-1.

orthophoto — An aerial photograph corrected so that it lines up with a map.

PACE, OCI — NASA’s 2024 ocean-colour satellite and its Ocean Color Instrument; family 1.gf’s **pace4k** store holds its daily 4 km bins.

Parquet — A columnar table file format, the form GBIF publishes its snapshots in.

phase A, B, C — The three build waves of each family: A cheap and first, B the terabyte stores, C decisions taken after A and B have verdicts.

pheno500 — Family 1.0.tf target store of the yearly leaf calendar at 500 m: the dates of green-up, peak, senescence and dormancy.

phenology — The calendar of a plant’s year — when it leafs out, peaks, dies back and goes dormant.

platforms.json — A file beside each station store giving every station’s identifier, elevation and the source’s metadata.

polarisation (HH, HV, VH, VV; dual, quad, compact) — The orientation of a radar’s transmitted and received waves (H horizontal, V vertical); dual-pol records two combinations, quad-pol all four, compact-pol a circular mix.

probe — A one-month (or one-zone) test run of a store’s real fetch code, measuring rows, bytes and speed before any full build.

public track, private track — Where a store lives: the public Hugging Face repository that anyone can read, or a private one used only by this project because the source’s licence forbids republishing.

QA, QC (qc) — Quality assurance and quality control: the producer’s flags saying how far a value can be trusted, kept as a channel or in the qc column.

RADD — Radar for Detecting Deforestation, Wageningen’s Sentinel-1 forest-loss alerts.

radiosonde — An instrument package on a weather balloon that measures temperature, humidity and wind as it rises.

read-out — A quantity the model is asked to predict or report, such as the Niño-3.4 index or next year’s forest loss.

reanalysis — A weather or ocean model run over past decades and continually corrected towards observations, giving complete fields that are partly model.

refl05 — Family 1.0.tf store of daily 0.05° land-surface reflectance.

registry — The JSON file listing every store of a family with its tier, location, channels, checksums and licence; the one file a reader opens first.

RG (Roemmich–Gilson pressures) — The 16 standard depths (2.5 to 2,000 dbar) of the Roemmich–Gilson Argo climatology, used as the vertical axis of every profile store.

RH25, RH50, RH75, RH90, RH98 — The heights below which 25, 50, 75, 90 and 98 % of a laser footprint’s returned energy lies; RH98 is about the canopy top.

river — Family 1.0.tf store of public daily river discharge (USGS and Caravan).

river_grdc — Private family 1.0.tf store of GRDC daily river discharge.

S3 (s3://) — Amazon’s object-storage service and its address form; many archives serve files through it.

S5P-PAL, TROPISIF — The ESA service that publishes extra Sentinel-5P products, and its fluorescence product, the open counterpart of the Caltech archive.

SANHO — The South African Navy Hydrographic Office.

ScanSAR — A wide-swath radar mode that trades resolution for coverage.

schema 2, schema 3 — The column layout of a point store: schema 3 widens the time column to 64 bits so records before 1914 fit.

seed — One independent training run; three seeds means the same test run three times with different random starts.

Sentinel-1, -2, -3, -5P, -6A (S1A, S1C, S1D, S2A, S2B, S2C) — The EU Copernicus satellites: C-band radar, 10 m optical, ocean colour and temperature, trace gases, and sea-level altimetry; the letters are individual satellites of a series.

shard, tile; shard_index.npy, tile_grid.json, n_tiles — In the G-fine layout, one compressed file per group and bin, cut into 256×256 -pixel tiles; the index gives each tile's byte position and the grid file each tile's corners.

sif — Family 1.0.tf store of solar-induced chlorophyll fluorescence soundings from TROPOMI and from OCO-2 and OCO-3.

SIF (solar-induced chlorophyll fluorescence) — The faint red light a leaf re-emits while photosynthesising; because the light is emitted by the process itself, it is the closest thing to a direct measurement of photosynthesis from orbit.

SLC, GRD, GRDH, IW — Sentinel-1 product forms: single-look complex (keeps the radar phase), ground-range detected (intensity only; H for high resolution), from the Interferometric Wide-swath mode.

snow05 — Family 1.0.tf store of daily 0.05° snow cover.

SST — Sea-surface temperature.

STAC — SpatioTemporal Asset Catalog, the standard in which producers list their scenes and files.

static.fine — Family 1.0.tf store of fixed fields: elevation and bathymetry, land-cover fractions, distance to coast.

store — One dataset of a family, published as a small set of arrays plus a description file.

store.json — The description file of a store: its arrays, their checksums, channels and counts.

swath — The strip of ground a satellite instrument sees on one pass.

SWISSIMAGE — swisstopo's 10 cm aerial orthophoto of Switzerland.

swisstopo — Switzerland's federal office of topography.

TAO — The tropical Pacific mooring array, held in family 10.1's **gtmba**.

tide — Family 1.0.tf store of public GESLA tide-gauge sea level.

tide_private — Private sibling of **tide** holding the gauges whose terms forbid republishing.

tier G-fine (sharded) — The layout family 1.gf adds for grids finer than 0.25° : each five-day bin is one compressed file cut into tiles, so a read fetches only the tiles it needs.

tier G, P, T — Family 10's three storage forms: G, gridded fields read one time step at a time; P, points, profiles and tracks read by nearest-neighbour search; T, image tiles held as a catalogue plus codec tokens.

tile tokens — The codec's vectors for each image tile, a later store sized by E-078c.

TMAX, TMIN, PRCP, SNOW, SNWD — GHCN-Daily element codes: maximum and minimum temperature, precipitation, snowfall and snow depth.

token — The unit the model reads: a value, its offset in space and time from the anchor, and its footprint.

TROPOMI — The trace-gas imager on Sentinel-5P, measuring NO_2 , CO and CH_4 (nitrogen dioxide, carbon monoxide, methane) — and, through the Caltech and S5P-PAL retrievals, solar-induced chlorophyll fluorescence at 740 nm.

T, S, u, v — Temperature, salinity, and the east and north components of velocity or wind.

TU Wien — The Technical University of Vienna, which hosts ISMN.

UCSB CHC — The Climate Hazards Center at the University of California, Santa Barbara, producer of CHIRPS.

UCSD — The University of California, San Diego, host of HFRNet's archive.

UHSLC — The University of Hawaii Sea Level Center.

UK, US, EU — United Kingdom, United States, European Union.

UMD — The University of Maryland, producer of GLAD and the Hansen forest-change map.

URL — A web address.

USDA — The US Department of Agriculture.

USGS — The US Geological Survey.

UTC — Coordinated Universal Time, the time zone of weather observations.

verdict — Registry field holding a store's falsifier result, empty until the test has run.

VIIRS; S-NPP, NOAA-20, NOAA-21 — The Visible Infrared Imaging Radiometer Suite, MODIS's successor, on the Suomi-NPP (ending 2026-11), NOAA-20 and NOAA-21 satellites; I-band is its 375 m channel set.

VNP02IMG, VJ102IMG, VJ109CMG, VJ209CMG, VNP10C1, VJ110C1, VJ121C1, VJ221C1 — VIIRS product names (VNP for Suomi-NPP, VJ1 for NOAA-20, VJ2 for NOAA-21): calibrated imagery, daily reflectance, snow cover and land-surface temperature.

WIND_GLO_PHY_L3_MY_012_005, WAVE_GLO_PHY_SWH_L3_MY_014_005, 009_103 — Copernicus Marine product identifiers: gridded scatterometer wind, along-track wave height, and GlobColour 300 m ocean colour.

WorldCereal — ESA's global crop-type map.

WorldCover — ESA's 10 m global land-cover map (2020 and 2021).

WSF — The World Settlement Footprint of built-up areas, yearly since 1985.

WUR — Wageningen University & Research, producer of RADD.